Upset Losses and Political Attitudes: Evidence from Chinese Football

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Abstract

Do incidental emotional shocks affect political attitudes toward state authority? We link Chinese Super League football results to respondents' interview dates in the Chinese General Social Survey (2015, 2018, 2021, and 2023) and exploit within-province-year variation in exposure to unexpected match outcomes, classified against pre-match betting odds. Upset losses by a respondent's provincial team increase expressed acceptance of government intervention against public criticism; the effect fades over the week after the match. The response is most pronounced in attitudes toward public criticism: parallel items on government interference in private life, social trust, and happiness show no comparable movement. These findings are consistent with transient negative emotions increasing acceptance of government control over public criticism, a reversal of the same-valence pattern documented in the political-spillover literature.

Keywords: incidental emotions, political attitudes, compensatory control, football, China

JEL Classification: D91, P20, Z20

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1 Introduction

When a sports team loses, the consequences can reach well beyond the stadium: if the team was favored to win, family violence rises and juvenile sentences lengthen (Card and Dahl, 2011; Eren and Mocan, 2018); national stock markets dip the next day (Edmans et al., 2007); even a narrow defeat can erode trust in political leaders (Kikuta and Uesugi, 2023). If a lost football match can shake citizens' trust in those who govern them, can it also change how much authority they are willing to grant the state? Acceptance of government control over political expression is central to any political order, yet little is known about what shapes it beyond the state's own conduct. China offers a natural setting for this question: the State Council has made football's development a national project (General Office of the State Council, 2015), and the professional league's clubs are widely followed and closely identified with their home provinces. We ask whether an unexpected loss by a province's football team, an emotional shock with no connection to politics, shifts citizens' acceptance of government control over public criticism.

Behind the increase in family violence and the longer juvenile sentences lies a common trigger: not the loss itself, but the gap between the realized and the expected outcome (Kőszegi and Rabin, 2006; Card and Dahl, 2011). The same reference-point logic has been extended to consumer spending and tipping, where unexpected outcomes move both (Cai and Li, 2026; Ge, 2018). Where these shocks reach political attitudes and behavior, the documented response has so far matched the valence of the shock (Healy et al., 2010; Kikuta and Uesugi, 2023). Whether a negative shock of this kind can instead *increase* acceptance of state authority, reversing the same-valence pattern, has not (to our knowledge) been documented.

We link Chinese Super League (CSL) match results to individual interview dates in four waves of the Chinese General Social Survey (CGSS) between 2015 and 2023. Following Card and Dahl (2011) and Ivandić et al. (2024), we classify these outcomes relative to pre-match betting odds, so that an upset

loss is a defeat in a match the bookmakers expected the team to win. Identification comes from timing: an upset is, by construction, the unanticipated part of the result, so whether a respondent happens to be interviewed just after one is plausibly as good as random. We therefore compare respondents in the same province and survey year whose interviews fall shortly after an unexpected defeat with those interviewed at other times; province-by-year fixed effects absorb conditions common to a province in a given year.

Upset losses reduce support for government non-interference with public criticism by 0.293 points on a five-point scale ($p < 0.01$) within a one-day window, an effect equivalent to 0.26 standard deviations. In other words, respondents interviewed the day after their provincial team suffers an unexpected defeat are more willing to accept government intervention when it is publicly criticized. The reduction fades to 0.222 points ($p < 0.01$) at three days and 0.148 points ($p < 0.05$) at seven days. This initial response followed by steady decay over the week is consistent with a transient emotional channel rather than a lasting attitude change. Close losses (defeats in matches the bookmakers rated as roughly even) and upset wins produce no comparable effects, and the response appears larger among respondents more likely to have seen the result. Upset losses are rare, and the effect is identified off a small number of treated provinces, so we complement conventional inference with randomization and wild-cluster bootstrap methods (Section 6). We rest the case not on any single test but on the specificity of the response: to surprise, to one political domain, and to more-exposed respondents.

The response is also not a diffuse bad mood. A general souring of mood would spill over into whatever judgment respondents are asked to make (Schwarz and Clore, 1983). Yet the observed shift is most pronounced in attitudes toward public criticism of the government. Nothing comparable appears for two companion items concerning government non-interference in childbearing or in choices of work and residence, nor does the shift extend to generalized social trust or self-reported happiness. One plausible interpretation of this domain-specific response is a reaction to threatened control. An upset loss is sudden,

unexpected, and beyond the fan's control; supporters may feel anxiety, frustration, or anger at players, coaches, or referees (Smith and Ellsworth, 1985; Lerner and Keltner, 2001), but none of these targets implicates the government. When personal control is threatened, individuals may restore a sense of order by bolstering the external systems that provide it, the government among them, a response known as compensatory control (Kay et al., 2008, 2009). Among the domains we examine, public criticism of the government most directly challenges the authority of this order-providing institution, which may explain why it is the only domain in which attitudes respond to an upset loss.

We contribute to two literatures. First, to the literature on incidental emotional shocks and their political spillovers, we add the missing direction: a negative shock that makes citizens *more*, not less, likely to accept state authority. In the political cases documented so far, the direction of the response has generally followed the valence of the sporting shock: pre-election wins raise incumbent vote shares (Healy et al., 2010), and close losses lower trust in national leaders (Kikuta and Uesugi, 2023). To our knowledge, we provide the first field evidence of the opposite pattern: an incidental negative shock, in which the state plays no part, increases expressed acceptance of government control over public criticism. This finding does more than add another instance of political spillover from sports. It indicates that the effect of a negative emotion on political attitudes depends less on the valence of the shock itself than on the attitude it meets, consistent with evidence that induced negative affect elsewhere shifts policy views without raising acceptance of authority (Algan et al., 2026). Its direction is also the one compensatory-control theory predicts for an outcome that concerns social order.

Second, to the study of political attitudes in China, we add a determinant that lies outside the government's own conduct. Existing work examines how citizens' trust in and support for the state respond to government performance and political information (Fang et al., 2025), how the state manages political expression (King et al., 2013), and how broadly favorable public opinion toward the government moves

with material conditions (Cunningham et al., 2020). We show that attitudes toward government control over public criticism also respond to incidental emotional cues orthogonal to what the state itself has done. The demand side of political control—the degree of authority citizens are willing to grant the state—may therefore be more malleable in the short run than accounts centered on instrumental evaluation or stable cultural values would suggest. A practical corollary is that short-run variation in measured political attitudes may reflect not only what the government has done, but also when citizens happen to be surveyed.

The remainder of the paper proceeds as follows. Section 2 discusses background and motivation. Section 3 describes the research design. Section 4 presents the main results and diagnostic evidence. Section 5 examines identification through event-study and balance tests. Section 6 reports robustness and inference. Section 7 concludes.

2 Background and Motivation

Three literatures jointly motivate our test and make it possible: sporting upsets provide credible incidental emotional shocks; the documented political responses to such shocks generally mirror their valence; and the psychology of threatened control identifies the conditions under which that pattern should reverse. Our design belongs to a lineage that treats sporting outcomes as incidental emotional shocks, isolated not by the result itself but by an expectation benchmark (Kőszegi and Rabin, 2006): an upset is defined against point spreads (Card and Dahl, 2011; Eren and Mocan, 2018; Cai and Li, 2026), betting odds (Ivandić et al., 2024; Ge, 2018), or rankings (Rees and Schnepel, 2009). The emotional cue is therefore orthogonal to whatever the benchmark already prices, including team quality, form, and the stakes of the match. In US family violence and juvenile sentencing data, the response follows the valence of the surprise, and negative surprises dominate (Card and Dahl, 2011; Cardazzi et al., 2024; Eren and

Mocan, 2018); asset prices fall after losses irrespective of pre-match expectations (Edmans et al., 2007); the exceptions are themselves informative, with upset losses raising retail-therapy spending (Cai and Li, 2026) and tips responding to surprise wins alone (Ge, 2018). Already within these behavioral domains, the direction of the response is not fixed by the valence of the shock alone.

The lineage rests on an individual-level microfoundation, and it has a known scope condition. Kessler et al. (2022) supply the former, instrumenting within-person happiness with live game events in a sports-bar experiment and moving incentivized decisions over charitable giving and trust.[1] German league evidence raises the scope question directly: odds-based upsets do not drive match-day violence, which tracks rivalry and group identity instead (Andres et al., 2023); whether surprise operates in a given setting is thus an empirical question. Table 1 organizes this evidence by shock, identification, and signed response.

Because upsets are rare events in observational data, the literature polices itself against false positives. Fowler and Montagnes (2015) reexamine the electoral result of Healy et al. (2010) and argue it is spurious: the estimate is no stronger where college football commands more attention, is undiminished where the mechanism predicts it should weaken (with no incumbent on the ballot, or outside the team's home county), and has no analogue in the far more popular NFL. We read the episode as setting the evidentiary bar for this design family rather than discrediting it. It asks that shocks be classified against market-priced expectations, that placebo outcomes bound what the shock can move, and that inference respect the small number of treated units; Sections 3, 4, and 6 implement each of these standards. Replication across settings matters for the same reason, and the same-valence political spillover does recur: national-team victories shift self-identification from the ethnic group toward the nation in sub-Saharan Africa (Depetris-Chauvin et al., 2020), and close losses by European clubs fielding African players de-

[1] Coviello et al. (2024) use the same class of instrument, local team results, for worker mood and find that *better* mood lowers output among fixed-wage call-center workers, a reminder that what a mood shift does depends on the margin it meets.

Table 1: Incidental emotional shocks and the direction of their spillovers.

Study	Shock (identification)	Response
Panel A: Behavioral and economic outcomes		
Card and Dahl (2011)	Upset NFL losses, US (point spread)	Family violence rises
Edmans et al. (2007)	International soccer losses (effect unrelated to pre-match expectations)	Stock returns fall
Eren and Mocan (2018)	Upset college football losses, US (point spread)	Juvenile sentences lengthen
Cai and Li (2026)	Unexpected NFL outcomes, US (point spread)	Spending rises after losses, falls after wins
Panel B: Political outcomes		
Healy et al. (2010)	Pre-election college football wins, US	Incumbent vote share rises
Depetris-Chauvin et al. (2020)	National-team victories, sub-Saharan Africa (interview timing; close qualification)	Ethnic identification falls; civil conflict falls
Kikuta and Uesugi (2023)	Close losses, European clubs fielding African players (interview timing; odds-conditioned)	Trust in leaders falls; peaceful protest rises
Guo et al. (2025)	Female Olympic gold medals, China (DiD across CGSS waves)	Desire for daughters rises
Chen and MacDonald (2020)	Broadcast of athletic achievements, China (randomized)	Local-government satisfaction and compliance rise
Panel C: Non-sports incidental affect		
Algan et al. (2026)	Induced negative emotions, US (survey experiments)	Protectionism and redistribution support rises; populist attitudes unchanged
Fochmann et al. (2025)	Induced positive emotions, Germany (laboratory)	Tax compliance falls
Meier et al. (2019)	Voting-weekend rainfall, Switzerland	Votes for political change fall

Notes: Each row is a representative study; where a parenthetical appears, it states the device isolating the unexpected or exogenous component of the shock. Related studies that replicate or qualify these designs are discussed in the text.

press trust in national leaders across Africa (Kikuta and Uesugi, 2023). Where the design family reaches political outcomes and clears this bar, the response has so far always shared the sign of the shock.

Evidence from China bears on each element of our design, but no study runs the test itself. Emotional cues from sports do shift attitudes in China: Chen and MacDonald (2020) randomize exposure to a broadcast of Chinese athletic achievements and find higher satisfaction with and compliance toward local government. Using the CGSS as we do, Guo et al. (2025) show that the Rio 2016 gold medals of female champions raised the desire for daughters in their home provinces across survey waves. Both studies examine favorable events. For a negative cue, the closest natural experiment is the Wenchuan earthquake: political trust rises among respondents interviewed just after the quake, and the rise dissipates within weeks (You et al., 2020). That event, however, bundles the shock with the state's own highly visible relief mobilization and media rally, so it captures (at least in part) citizens' evaluations of government in action. Isolating incidental emotion requires a shock in which the state plays no part, which club football provides. On the outcome itself, attitudes toward the state in China are often modeled as instrumental evaluations of government performance (Fang et al., 2025; Cunningham et al., 2020), while direct questions on sensitive items may overstate support because of self-censorship (Robinson and Tannenberg, 2019), a measurement concern that a shock-based design can difference out insofar as reporting incentives are stable within the exposure window (Section 3). What remains unanswered is whether an emotional cue orthogonal to government action moves expression-related attitudes at all.

Returning to the full literature, two gaps follow from Table 1. First, every political spillover from a sporting shock, Panel B in full, runs in the same direction as the shock; the departures visible elsewhere in the table lie outside the political panel: retail-therapy spending after upset losses in Panel A and the non-sports patterns of Panel C. Second, none of the outcomes concern acceptance of state limits on political expression. That the valence mapping is outcome-dependent is precisely what the non-sports evidence

shows: induced negative emotions raise support for protectionist and redistributive policies in the United States without hardening attitudes toward authority (Algan et al., 2026), incidental positive emotions lower tax compliance (Fochmann et al., 2025), and rain-induced negative affect shifts referendum votes toward the status quo (Meier et al., 2019). In these studies, the direction of the spillover tracks what the emotion disposes people to do (seek protection, hold to the familiar, or let compliance slip), rather than the valence of its trigger. No row of Table 1 pairs a negative incidental shock with rising acceptance of state authority: to our knowledge, that combination, acceptance of state limits on public criticism after an incidental negative shock, remains undocumented.

Appraisal theory explains why a reversal should emerge here: incidental emotions carry their appraisal tendencies, not merely their valence, into unrelated judgments (Lerner and Keltner, 2001). An upset loss is sudden, expectation-violating, and beyond control, and the agents it offers for blame—players, coaches, or referees—lie outside the state. This profile is associated with appraisals of uncertainty and low control (Smith and Ellsworth, 1985); anger, organized around perceived unfairness and an identifiable target of blame, is the configuration that theories of political unrest link to anti-government action (Passarelli and Tabellini, 2017), and after a defeat that target is the team, not the government. Compensatory-control theory predicts how people respond when their sense of personal control is threatened: they seek to restore a sense of order by bolstering external systems that can provide it, with greater support for government among the best-documented responses (Kay et al., 2008, 2009; Landau et al., 2015). Consistent with this account, uncertainty and threat raise the appeal of strong, order-providing authority (Kakkar and Sivanathan, 2017), with perceived threat activating authoritarian predispositions (Feldman and Stenner, 1997). For this compensatory turn to favor the state, the state must be regarded as a credible provider of order. That premise is plausibly met in China, where approval of the central government is persistently high (Cunningham et al., 2020), and local governments respond to citizen

demands (Chen et al., 2016), complementing institutional accounts of regime resilience (Nathan, 2003). The resulting prediction is precise: heightened concern for order should be directed specifically at the behavior perceived to threaten order, public criticism of the government, rather than at government control in general. The placebo outcomes in Section 4 are built to detect exactly this specificity. No existing study has run this test on expression-related attitudes with an incidental shock in China; addressing this gap is the central contribution of our paper.

3 Research Design

3.1 Institutional Setting: The Chinese Super League

The Chinese Super League (CSL) is the top tier of professional football in China. Founded in 2004, the league comprises 16 clubs, each based in a provincial capital or major city and closely identified with its home province. In March 2015, the General Office of the State Council issued the Overall Plan for Chinese Football Reform and Development, a national reform blueprint for the game (General Office of the State Council, 2015; Cockayne et al., 2022). Much of our sample falls in the league's *jinyuan* ("money football") boom years, when clubs backed by property developers and state-linked owners recruited established international stars: they broke the Chinese transfer record five times in 2016 alone (Church, 2016), with Chelsea's Oscar arriving for 60 million pounds (Twomey, 2016). The boom showed in the stands and on screens: by the 2015 season the league averaged above 21,800 spectators per match, within a few hundred of Serie A and Ligue 1 (Hennelly, 2016), and its five-year broadcast rights for 2016–2020 sold for a record 8 billion yuan, about 1.25 billion USD (Xinhua, 2015). The combination of strong provincial identification with local clubs, star players, and broad media reach makes CSL outcomes a plausible source of short-lived emotional shocks for the general provincial population.

3.2 Data

Football data Following Card and Dahl (2011) and Ivandić et al. (2024), we use betting markets to measure pre-match expectations and classify match outcomes into emotional-cue categories. Pre-match odds define the benchmark against which fans evaluate realized outcomes (Kőszegi and Rabin, 2006), so a defeat when the team was favored generates a larger negative response than one that was expected (Card and Dahl, 2011). We collect match-level data for all CSL games from 2014 to 2025 from `www.football-data.co.uk` (2026). The data provide the match date, home and away teams, full-time result (home win, draw, or away win), and pre-match decimal odds for each of the three outcomes. We additionally scrape match-level home attendance and club net transfer spending for all CSL clubs from Transfermarkt (2026), which enter the robustness analysis as controls (Figure 6).

We convert odds into implied probabilities and remove the bookmaker margin by normalizing inverse odds so that $p_H + p_D + p_A = 1$. Using the team's implied win probability p_{win}, we classify each match from each team's perspective as a *predicted win* ($p_{\text{win}} \geq 0.55$), *predicted loss* ($p_{\text{win}} < 0.45$), or *predicted close* (otherwise). These thresholds follow Ivandić et al. (2024), who apply the same cutoffs to matches of Manchester United and Manchester City.[2]

Combining these predicted-outcome bins with realized results yields six mutually exclusive categories: *upset loss* (expected win, actual loss), *close loss* (close match, actual loss), *predicted loss* (expected loss, actual loss), and the corresponding win-side categories *upset win*, *close win*, and *predicted win*. Throughout, "close" refers to the pre-match expectation rather than the final scoreline: a close loss is a defeat, by whatever margin, in a match the market priced as roughly even; the category labels follow Card and Dahl (2011) and Ivandić et al. (2024). Draws form a separate category and do not enter

[2] Card and Dahl (2011) classify NFL games using a point-spread threshold of 4 points. Because football (soccer) betting markets quote three-way odds rather than point spreads, we adopt the probability-based classification of Ivandić et al. (2024), which maps onto an analogous distinction between games where the team is a clear favorite, games that are roughly evenly matched, and games where the team is a clear underdog.

the win/loss classifications. An upset loss thus captures the negative surprise of a defeat when the team was favored, while a predicted loss carries little new information and should produce a weaker response (Kőszegi and Rabin, 2006; Card and Dahl, 2011).

Figure 1 plots the realized goal difference (home goals minus away goals) against the implied home-win probability for all CSL matches in our sample. The positive relationship validates betting odds as informative measures of pre-match expectations.

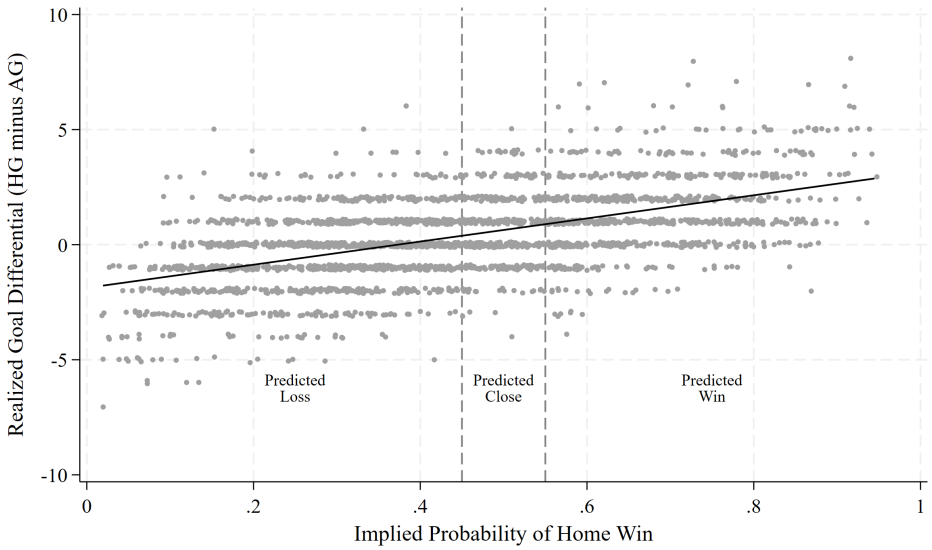


Figure 1: Betting market efficiency.

Notes: Each point is a Chinese Super League match, 2014 to 2025. The horizontal axis is the implied home-win probability from normalized betting odds (football-data, Pinnacle closing); the vertical axis is the realized goal difference (home minus away). The solid line is a linear fit; the vertical dashed lines mark the $p = 0.45$ and $p = 0.55$ thresholds separating predicted losses, close matches, and predicted wins.

Household survey data We draw individual-level data from the Chinese General Social Survey (CGSS), a nationally representative repeated cross-section administered by the National Survey Research Center at Renmin University of China (2025). The CGSS uses a multi-stage stratified probability sampling design covering most provinces in mainland China and records the exact date of each interview, which is essential for linking respondents to recent match outcomes in their province. We pool the 2015, 2018,

2021, and 2023 waves, which share a common set of attitudinal and demographic items.[3]

Respondents rate their agreement with the statement that "the government should not interfere when someone publicly criticizes it," on a scale from 1 (strongly disagree) to 5 (strongly agree), which we label *Non-interference: public criticism*. A respondent who fully agrees scores a 5, the strongest support for non-interference, so a negative treatment coefficient reflects greater acceptance of government intervention against public criticism. We interpret the item as measuring tolerance of state restrictions on public criticism, rather than as a broad evaluation of the government or of freedom of expression in the abstract. In the terminology of the political-tolerance literature, it is a tolerance-of-dissent item in which the acting party is the state rather than the respondent: the question is whether the government should refrain from acting against a critic, not whether the respondent would extend expressive rights to speech they dislike. Because the item does not specify a particular level of government, we interpret responses as referring to government authority in general. This construct connects to research on how the Chinese state manages political expression (King et al., 2013) and to the compensatory-control account we adopt, in which a threatened sense of personal control lowers willingness to tolerate criticism of the government (Kay et al., 2009). Direct questions on politically sensitive items can also overstate support through self-censorship (Robinson and Tannenberg, 2019). Our design mitigates this concern by exploiting the as-if-random timing of an emotional shock: any stable tendency to misreport should be differenced out across exposed and unexposed respondents and therefore is unlikely to generate a response confined to one item and one short window. A transient shift in self-censorship triggered by the shock itself would not be differenced out this way; we return to this in Section 7.

To gauge the specificity of this response, we also examine four placebo outcomes that should be unmoved if the effect is a domain-specific reaction to political criticism: two other government-non-

[3]We exclude the 2017 wave, which does not record interview dates and therefore cannot be linked to match outcomes.

interference items, and two general-mood measures.[4]

Individual-level controls include gender, age, Han ethnicity, years of education, marital status, Chinese Communist Party membership, agricultural *hukou* status, self-rated health, and the log of personal income. These variables capture the key demographic and socioeconomic characteristics that prior work has shown to predict political attitudes and subjective well-being in China (Fang et al., 2025; Zhang et al., 2025; Asadullah et al., 2018; Zhang and Awaworyi Churchill, 2020). We additionally control for television and internet use frequency (each on a 1–5 scale) to account for differential media exposure that might correlate with both awareness of match outcomes and attitudinal responses.

Province-level variables We do not include province-level covariates such as GDP per capita, population, or industrial structure in the main regressions, because the province-by-year fixed effects absorb all province-year variation, including any contemporaneous economic conditions that could correlate with both match outcomes and political attitudes.

3.3 Descriptive Statistics

Table 2 reports descriptive statistics for the estimation sample. On average, respondents score 2.55 on the five-point scale of support for government non-interference with public criticism, indicating that the typical response lies between "somewhat disagree" and "neither agree nor disagree." The football shock variables in Panel B show that upset losses are rare events in a one-day window (mean $= 0.002$), which motivates our use of short pre-interview windows spanning up to one week in the main analysis. Individual characteristics in Panel C are broadly consistent with the composition of the adult survey sample: 50% female, 92% Han, and roughly nine years of schooling on average.

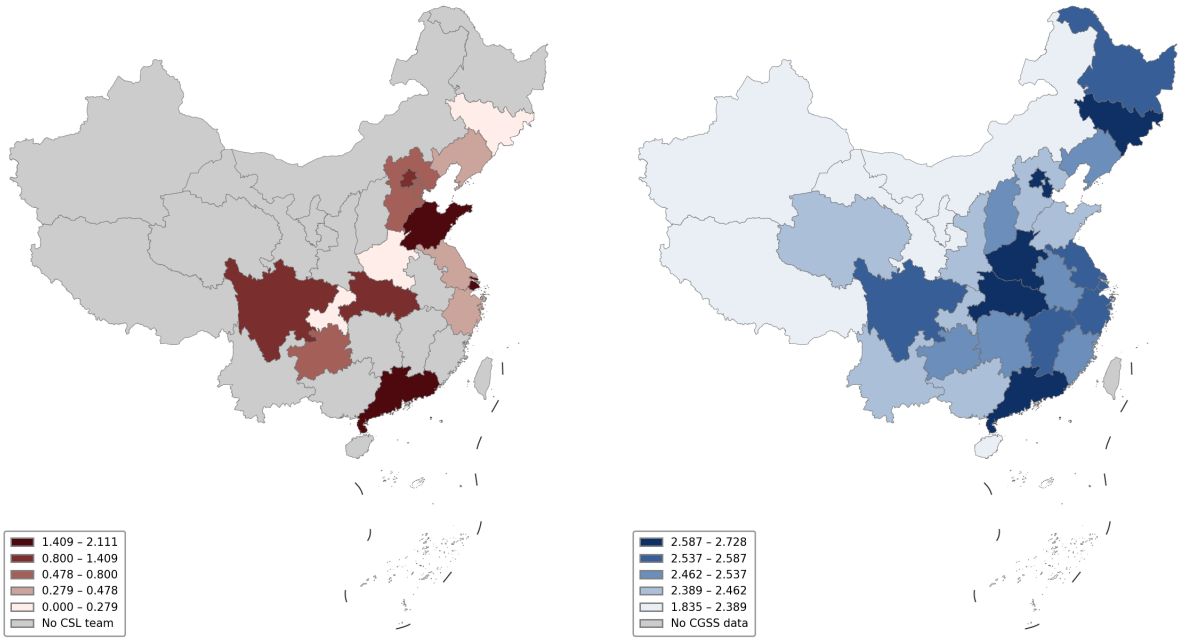
[4]The two additional government-non-interference items ask for agreement, on the same 1–5 scale (higher indicating greater support for non-interference), that "how many children to have is a personal matter, and the government should not interfere" and that "where to work and live is a personal freedom, and the government should not interfere." The two mood measures are generalized social trust and life happiness, each on a 1–5 scale on which higher values indicate, respectively, greater trust and greater happiness. All four are summarized in Table 2 and used in Figure 3.

Table 2: Summary statistics.

Variable	N	Mean	Std. Dev.	Min	Max
Panel A: Outcome variables					
Non-interference: public criticism (1–5)	35,103	2.551	1.126	1	5
Non-interference: childbearing (1–5)	34,846	2.856	1.222	1	5
Non-interference: work/residence (1–5)	34,845	3.736	1.105	1	5
Social trust (1–5)	34,998	3.531	0.986	1	5
Life happiness (1–5)	32,741	3.899	0.818	1	5
Panel B: Football shock variables (1-day window)					
Upset loss	35,103	0.002	0.040	0	1
Close loss	35,103	0.001	0.029	0	1
Upset win	35,103	0.012	0.109	0	1
Predicted win	35,103	0.017	0.132	0	2
Close match	35,103	0.009	0.092	0	1
Predicted loss	35,103	0.060	0.244	0	2
Panel C: Individual controls					
Female (=1)	35,103	0.497	0.500	0	1
Age (years)	35,103	48.819	17.110	18	118
Han ethnicity (=1)	35,103	0.923	0.267	0	1
Education (years)	35,103	9.443	4.580	0	19
Married (=1)	35,103	0.749	0.434	0	1
CCP member (=1)	35,103	0.116	0.320	0	1
Agricultural hukou (=1)	35,103	0.563	0.496	0	1
Self-rated health (1–5)	35,103	3.599	1.074	1	5
Log personal income	35,103	8.409	3.944	0	16
Panel D: Media controls					
TV use frequency (1–5)	35,103	3.482	1.183	1	5
Internet use frequency (1–5)	35,103	3.194	1.681	1	5

Notes: Estimation sample for the main regressions: CGSS respondents from the 2015, 2018, 2021, and 2023 waves with non-missing individual controls, media controls, and sampling weights ($N = 35{,}103$). Football shock variables are counts for the single day before the interview ($d - 1$). All statistics use CGSS sampling weights.

Figure 2 illustrates the geographic distribution of our key variables. Panel (a) shows the average number of upset losses per season in which the province had a CSL team; these provinces lie primarily in eastern and central China. Panel (b) shows the province-level mean of support for government non-interference with public criticism. The maps highlight two features of the empirical setting. First, there is substantial cross-province variation in the attitudinal outcome, which is absorbed by the province-by-year fixed effects. Second, the provinces most exposed to football shocks (e.g., Shandong, Guangdong, and Shanghai) span a range of attitudinal levels, mitigating concerns that the results are driven by a single type of province.



(a) Upset losses per year

(b) Non-interference: public criticism

Figure 2: Geographic variation in upset losses and support for government non-interference.

Notes: Province-level maps of China. Panel (a) shows the average number of upset losses (unexpected defeats based on pre-match betting odds) per season in which the province had a CSL team, 2015 to 2023. Panel (b) displays the province-level weighted mean of the *Non-interference: public criticism* item (1 to 5 scale; higher values indicate greater support for non-interference) from the CGSS. Gray provinces have no CSL team in Panel (a) and no CGSS data in Panel (b). The base map is the standard map of China (approval number GS(2024)0650) from the Ministry of Natural Resources' National Platform for Common Geospatial Information Services (Tianditu; Ministry of Natural Resources of the People's Republic of China, 2025); boundaries unmodified.

3.4 Empirical Strategy

We exploit quasi-random variation in the timing and outcomes of CSL matches relative to CGSS interview dates within the same province and year. Provinces with CSL teams differ systematically from those without, but the exact date on which a team plays, and whether it wins or loses relative to expectations, are plausibly orthogonal to the date on which a given respondent is interviewed, conditional on province-by-year fixed effects. This design draws on the logic of the Unexpected Event during Survey Design (UESD) framework (Muñoz et al., 2020), which exploits the as-if-random timing of an unexpected event relative to survey interviews; we implement it as the province-by-year design below. We estimate:

$$Y_{ipt} = \beta_1 \,\text{UpsetLoss}_{p,W} + \beta_2 \,\text{CloseLoss}_{p,W} + \beta_3 \,\text{UpsetWin}_{p,W} + \boldsymbol{\gamma}'\mathbf{C}_{p,W} + \boldsymbol{\delta}'\mathbf{X}_{ipt} + \alpha_{pt} + \varepsilon_{ipt}, \quad (1)$$

where Y_{ipt} is support for government non-interference with public criticism for individual i in province p interviewed in year t. The variables $\text{UpsetLoss}_{p,W}$, $\text{CloseLoss}_{p,W}$, and $\text{UpsetWin}_{p,W}$ count the number of upset losses, close losses, and upset wins experienced by the province's CSL team in the W-day window preceding the interview, specifically over the interval $[d - W,\, d - 1]$, where d is the interview date (we suppress d in the shock-count notation). We exclude the interview day itself to ensure that only pre-interview shocks enter the specification. The omitted category comprises respondents whose provincial team experienced none of the three shock outcomes during the window (its matches, if any, ended only in predicted wins, close wins, predicted losses, or draws). Following Card and Dahl (2011), we include predicted-outcome controls $\mathbf{C}_{p,W}$, consisting of counts of predicted wins, all close matches, and predicted losses within the same window. These controls ensure that the shock coefficients β_1 to β_3 capture the effect of deviations from expected outcomes rather than the match schedule itself. With total close matches held constant, β_2 captures the effect of a close loss relative to a close win.

The vector $\mathbf{X}_{ipt}$ includes individual-level controls: gender, age, Han ethnicity, years of education, marital status, CCP membership, agricultural *hukou* status, self-rated health, log personal income, and television and internet use frequency. The term α_{pt} denotes province-by-year fixed effects, which absorb all time-invariant province characteristics as well as province-specific annual factors that may correlate with team performance and respondents' attitudes. Identification therefore relies entirely on within-province-year variation: we compare respondents interviewed in the same province and survey year but on dates that differ in their exposure to recent unexpected match outcomes. Standard errors are clustered at the province level. All regressions are weighted using CGSS sampling weights.

The coefficient of primary interest is β_1. If an upset loss elicits negative arousal that increases demand for order and external control, we expect $\beta_1 < 0$: an upset loss should reduce support for the proposition that the government should not interfere with public criticism, i.e., increase acceptance of government intervention. The coefficient β_2 on close losses serves as a benchmark: these are losses from roughly evenly matched games that carry little surprise and should therefore produce a weaker emotional response. The coefficient β_3 on upset wins tests whether positive surprises generate comparable short-run spillovers.

We estimate equation (1) across three window widths, $W \in \{1, 3, 7\}$ days. Shorter windows ($W = 1, 3$) isolate acute responses to recent match outcomes, while the 7-day window allows us to examine whether these effects persist within the following week. The comparison across windows is informative about the nature of the effect: a sharp response concentrated in the shortest windows is consistent with a transient emotional cue rather than a durable change in attitudes (Card and Dahl, 2011).

Identifying assumption. The validity of our approach requires that, conditional on province-by-year fixed effects, the timing and outcomes of CSL matches are uncorrelated with unobserved determinants of individual attitudes. Two features of our setting support this assumption. First, the CSL schedule is

set by the Chinese Football Association before the season begins and is not influenced by provincial socioeconomic conditions or survey logistics. Second, CGSS interview dates are determined by fieldwork protocols, including interviewer availability, respondent scheduling, and sampling sequences, which are plausibly independent of match outcomes. We probe this assumption in several ways in Section 5: an event-study specification with placebo leads, which adds post-interview windows ($[d+1,\, d+3]$ and $[d+4,\, d+7]$) to binned pre-interview windows and in which significant effects of future match outcomes on current attitudes would indicate confounding pre-trends or spurious seasonality, and a balance comparison of respondents interviewed just before and just after an upset.

4 Results

4.1 Main Results

Table 3 reports estimates of equation (1) across three pre-interview windows. In the one-day window (column 1), an upset loss reduces support for non-interference by 0.293 points on the five-point scale ($p < 0.01$), equivalent to 0.26 standard deviations. The effect attenuates but remains significant at three days (-0.222, $p < 0.01$) and seven days (-0.148, $p < 0.05$). This steady decay pattern indicates a short-run emotional channel (Card and Dahl, 2011): an incidental-emotion spillover into unrelated judgments (Lerner et al., 2015) that fades over the week. The estimate is an intent-to-treat effect at the province level, averaging over all residents of the province. The pattern is unchanged under ordered logit and ordered probit (Appendix Table A4).

Two features temper how much weight the analytic p-values can carry. Because upset losses are rare, the number of treated clusters is small: two provinces are exposed within the one-day window and four within the three- and seven-day windows. With so few treated clusters, cluster-robust tests are known to over-reject severely, and the null-imposing Webb wild-cluster bootstrap we implement is known to

under-reject (MacKinnon and Webb, 2018, 2020); consistent with this, it returns conservative p-values of 0.15 to 0.23. MacKinnon and Webb (2020) find that, with few treated clusters, randomization inference is typically more reliable when based on t-statistics; Section 6.2 reports studentized permutation tests, and the upset-loss estimate clears the 5% level at three days under the baseline permutation scheme. We therefore read the analytic significance cautiously and rest the case on the broader pattern—the specificity to surprise, to the political domain, and to more-exposed respondents—rather than on any single test. The point estimate itself is stable, with no sign flips when any single province or event is dropped; Appendix Table A1 reports the wild-cluster bootstrap and the leave-one-out estimates.

Close losses show no comparable negative effect in any window. This contrast indicates that deviations from expectations, rather than losses per se, drive the result, consistent with reference-dependent utility (Kőszegi and Rabin, 2006). Upset wins are likewise insignificant, indicating that the spillover is asymmetric: negative surprises exert a stronger immediate effect than positive ones, in line with the well-documented negativity bias in affective responses (Baumeister et al., 2001).

Table 3: Effect of unexpected match outcomes on support for government non-interference.

	(1) Day 1	(2) Day 3	(3) Day 7
Upset loss	-0.293***	-0.222***	-0.148**
	(0.094)	(0.054)	(0.058)
Close loss	0.035	0.102	0.110*
	(0.091)	(0.136)	(0.059)
Upset win	-0.064	-0.059	-0.022
	(0.066)	(0.042)	(0.040)
Mean of dep. var.	2.551	2.551	2.551
Province × year FE	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes
Media controls	Yes	Yes	Yes
N	35,103	35,103	35,103
R^2	0.057	0.058	0.058

Notes: Dependent variable: *Non-interference: public criticism* (1–5, higher values indicating greater support for non-interference). Upset loss, close loss, and upset win are counts of each outcome type in the pre-interview window, classified from football-data betting odds (Pinnacle closing); the omitted category comprises windows with none of the three outcomes (matches, if any, ended only in predicted wins, close wins, predicted losses, or draws). All specifications include province × year fixed effects, individual controls, media controls, and Card predicted-category controls; pre-interview window $[d-W,\ d-1]$ with $W \in \{1, 3, 7\}$; province-clustered standard errors in parentheses; CGSS sampling weights ($N = 35{,}103$). $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$.

4.2 Interpretation and Diagnostic Evidence

The effect is domain-specific. Figure 3 plots the upset-loss coefficient, with 95% confidence intervals, for the main outcome and four placebo outcomes at each window. An upset loss reduces support for government non-interference with public criticism (reproducing the main estimate, -0.293 to -0.148 across windows), and no placebo outcome reproduces this profile: generalized social trust, life happiness, and non-interference in choices of work and residence are flat in every window, while non-interference in childbearing shows a weaker same-signed estimate that is marginal at seven days only (-0.136, $p = 0.054$), without the sharp one-day response that marks the main effect. A diffuse worsening of mood would move trust and happiness; a generalized rise in support for government control would move both companion items at once; neither pattern appears. This pattern instead fits the domain specificity of appraisal-driven emotions (Lerner and Keltner, 2001) and weighs against an account in which upset losses proxy for a broader provincial shock.

The effect is also concentrated among respondents more likely to have seen the result. An unexpected defeat of the provincial team is salient news that circulates widely online and in conversation, so the relevant margin is exposure rather than fandom. When we interact the shock with internet use, the closest CGSS proxy for connectedness to those information flows, the response is larger among high-internet respondents, consistent with an information-exposure channel (Appendix Table A3). This heterogeneity shares the few-treated-cluster limitation of the design and is read as directional rather than sharply estimated.

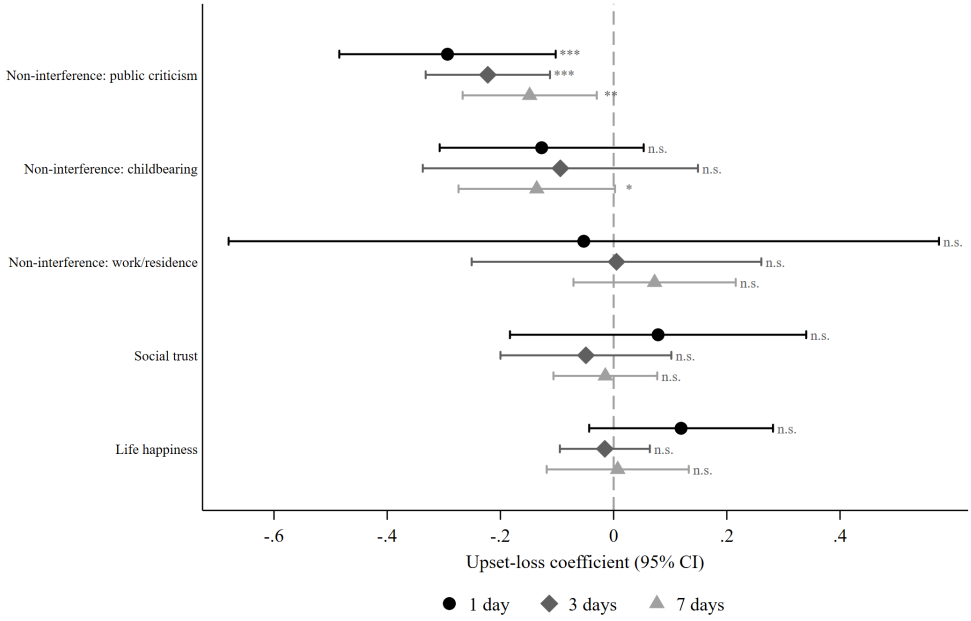


Figure 3: Domain specificity of the upset-loss effect.

Notes: The five outcomes are CGSS items, each on a 1–5 scale (higher values indicating more support for non-interference, or greater trust or happiness): three *Non-interference* items (public criticism, the main outcome; childbearing; work/residence) and two mood measures (social trust, life happiness). Each marker is the upset-loss coefficient, with its 95% confidence interval, from a separate regression per outcome and window under the Table 3 specification. Confidence intervals use province-clustered standard errors; CGSS weights.

5 Identification

This section probes the identifying assumption that exposure to an upset is unrelated to who is interviewed, through the timing of the effect and the balance of predetermined characteristics.

5.1 Event-Study Evidence

Figure 4 replaces the cumulative windows with binned counts of upset losses occurring 1 to 3 and 4 to 7 days before the interview, alongside placebo bins for losses 1 to 3 and 4 to 7 days after, which should be zero under the identifying assumption. The 1-to-3-day pre-interview bin shows a significant negative effect (-0.194, $p = 0.001$), while the 4-to-7-day bin remains negative but is not statistically significant. Both post-interview placebo bins are indistinguishable from zero, providing reassurance against pre-trends or reverse causality.

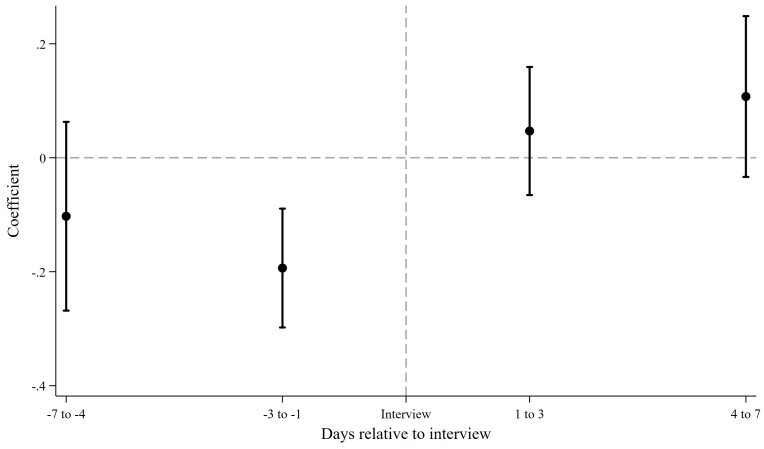


Figure 4: Event-study evidence: binned leads and lags of upset losses.

Notes: Dependent variable: *Non-interference: public criticism* (1–5, higher values indicating greater support for non-interference). Each marker is the coefficient, with its 95% confidence interval, on the number of upset losses in a bin relative to the interview date: negative event times are pre-interview windows (losses 1–3 and 4–7 days before the interview), positive event times are post-interview placebo windows (losses 1–3 and 4–7 days after). The specification includes province $\times$ year fixed effects and individual and media controls; CGSS weights ($N = 35{,}103$). Bars are 95% confidence intervals with province-clustered standard errors.

5.2 Balance and Composition

If exposure to an upset is unrelated to who is interviewed, respondents reached just before and just after an upset should be comparable. Figure 5 reports the before-versus-after comparison of Muñoz et al. (2020), plotting for each predetermined characteristic the standardized difference between respondents interviewed just after an upset (treated) and just before it (control, not yet exposed) within the three-day window, with its 95% confidence interval. Five of the seven characteristics are balanced (their intervals cross zero), the interval for CCP membership just touches zero, and a joint test does not reject balance ($F = 1.50$, $p = 0.374$ at three days, and $F = 0.52$, $p = 0.697$ at seven days). The one clear exception is age: post-upset respondents are somewhat older. This gradient reflects older respondents being reached later in the fieldwork period rather than the upset itself; it is indistinguishable from zero at seven days, and age is among the controls in every specification.

Re-estimating the effect on this strict local sample reproduces the main result. Comparing support for non-interference just after versus just before an upset yields -0.222 at three days, matching the -0.222 in Table 3, so the estimate is not an artifact of the broader within-province-year control group on which the main specification relies.

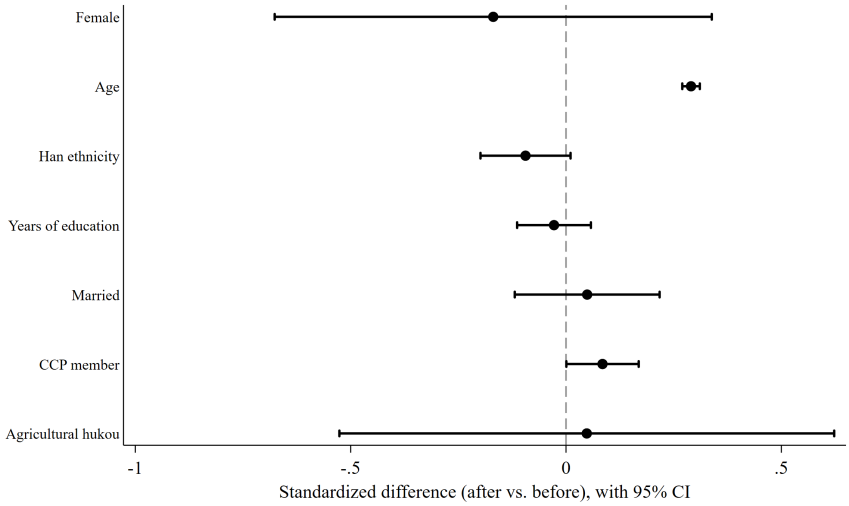


Figure 5: Balance of predetermined characteristics before and after an upset.

Notes: Canonical UESD design (Muñoz et al., 2020): within the three-day window around each upset, respondents interviewed just before it (not yet exposed) are the control and those just after are treated. Each marker is the standardized difference in a predetermined characteristic (after minus before, in standard-deviation units), with its 95% confidence interval (± 1.96 province-clustered standard errors), from a regression on the after indicator. Province $\times$ year fixed effects, CGSS weights.

6 Robustness and Inference

This section shows that the main estimate is stable across a wide range of alternative treatment definitions, control sets, samples, and fixed-effects structures, and probes inference with procedures designed for few treated clusters.

6.1 Robustness

Table 4 varies the definition of the treatment. Panel A redefines it as a single count or measure: an undifferentiated count of every loss produces no negative effect, and if anything a marginally positive one-day coefficient (0.128, $p < 0.10$), confirming that it is the surprise of an upset rather than losing as such that moves attitudes, while a favorite-did-not-win count and a continuous measure of favorite underperformance both reproduce the effect. Panel B raises the favorite threshold from 0.55 to 0.60 or 0.65, leaving the effect negative and mostly larger in magnitude, while the close-loss and upset-win

placebos remain mostly small and insignificant; the scattered exceptions are sign-inconsistent, led by a positive close-loss coefficient (0.258) in the one-day window at the 0.60 threshold, and no placebo reproduces the negative, decaying profile of the upset-loss effect. Figure 6 instead varies the controls and fixed effects: adding flexible interview-timing controls (day-of-week and month fixed effects and a holiday indicator), adding controls for the losing team's quality, form, ranking, home advantage, match importance, and finances, replacing the province-by-year fixed effects with separate province and year fixed effects plus province-level controls (National Bureau of Statistics of China, 2026), and controlling for the exact home-match attendance respondents were exposed to in the days before the interview; each variation leaves the coefficient essentially unchanged. The effect also reflects the occurrence of an upset more than its margin of defeat (Appendix Table A2). Because the treatment is identified off few treated provinces, these checks speak to the stability of the point estimate rather than to its precision; we examine inference separately through the randomization inference reported below and the wild-cluster bootstrap and leave-out estimates in Appendix Table A1.

Table 4: Robustness to alternative treatment definitions.

Panel A: Alternative treatment definitions (each column a separate single-treatment regression)									
	Any loss (placebo)			Favorite did not win			Continuous		
	Day 1	Day 3	Day 7	Day 1	Day 3	Day 7	Day 1	Day 3	Day 7
Redefined treatment	0.128*	0.058	0.038	-0.454***	-0.199***	-0.099***	-0.445***	-0.237***	-0.133***
	(0.070)	(0.045)	(0.037)	(0.149)	(0.066)	(0.021)	(0.110)	(0.077)	(0.029)
N	35,103	35,103	35,103	35,103	35,103	35,103	35,103	35,103	35,103
R^2	0.057	0.057	0.057	0.058	0.058	0.058	0.058	0.057	0.057
Panel B: Higher upset thresholds (each column a single joint regression)									
	≥ 0.55 (baseline)			≥ 0.60			≥ 0.65		
	Day 1	Day 3	Day 7	Day 1	Day 3	Day 7	Day 1	Day 3	Day 7
Upset loss	-0.293***	-0.222***	-0.148**	-0.404***	-0.270***	-0.220***	-0.368***	-0.133	-0.181*
	(0.094)	(0.054)	(0.058)	(0.066)	(0.055)	(0.067)	(0.071)	(0.079)	(0.090)
Close loss	0.035	0.102	0.110*	0.258***	0.085	0.065	0.110*	-0.032	-0.014
	(0.091)	(0.136)	(0.059)	(0.065)	(0.071)	(0.060)	(0.060)	(0.049)	(0.044)
Upset win	-0.064	-0.059	-0.022	-0.040	-0.042	-0.035	-0.074	-0.077**	-0.028
	(0.066)	(0.042)	(0.040)	(0.055)	(0.041)	(0.045)	(0.058)	(0.035)	(0.046)
N	35,103	35,103	35,103	35,103	35,103	35,103	35,103	35,103	35,103
R^2	0.057	0.058	0.058	0.058	0.058	0.058	0.057	0.057	0.057

Notes: Dependent variable: *Non-interference: public criticism* (1–5, higher values indicating greater support for non-interference). Panel A replaces the upset-loss count with three single-treatment definitions: an any-loss count (every loss, odds ignored), a favorite-did-not-win count (a favorite with implied win probability at least 0.55 that lost or drew), and a continuous favorite-underperformance measure (the gap between the favorite's implied win probability and the realized result, on a probability scale). Panel B re-estimates the joint specification at favorite (underdog) thresholds of 0.55 (0.45; baseline, as in Table 3), 0.60 (0.40), and 0.65 (0.35). All specifications include province $\times$ year fixed effects, individual controls, media controls, and Card predicted-category controls, except the continuous-measure columns, which instead control for the number of matches in the window; pre-interview window $[d - W,\, d - 1]$ with $W \in \{1, 3, 7\}$; province-clustered standard errors in parentheses; CGSS sampling weights ($N = 35{,}103$). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

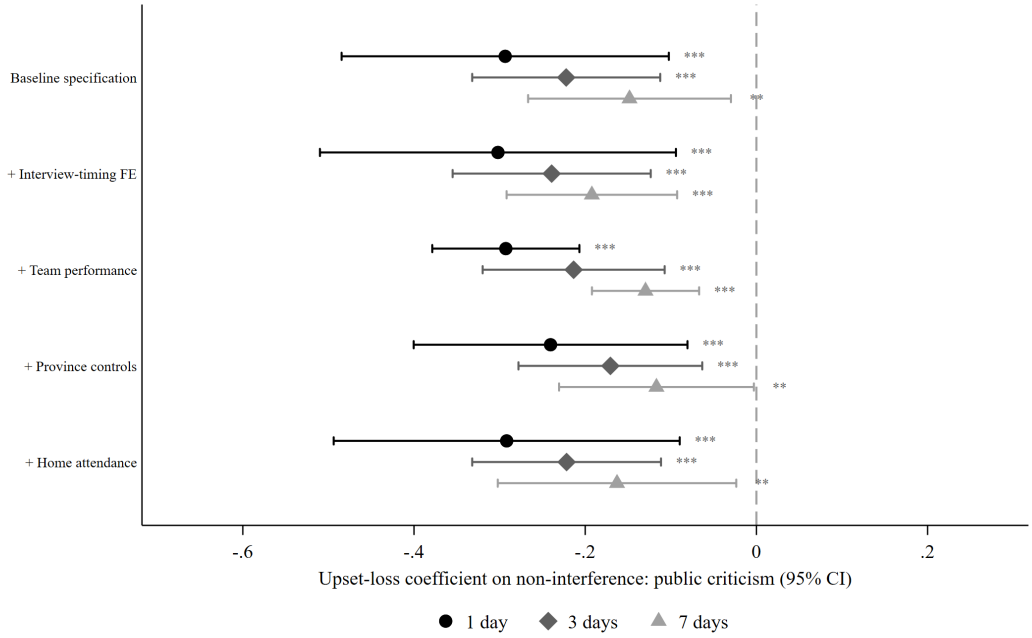


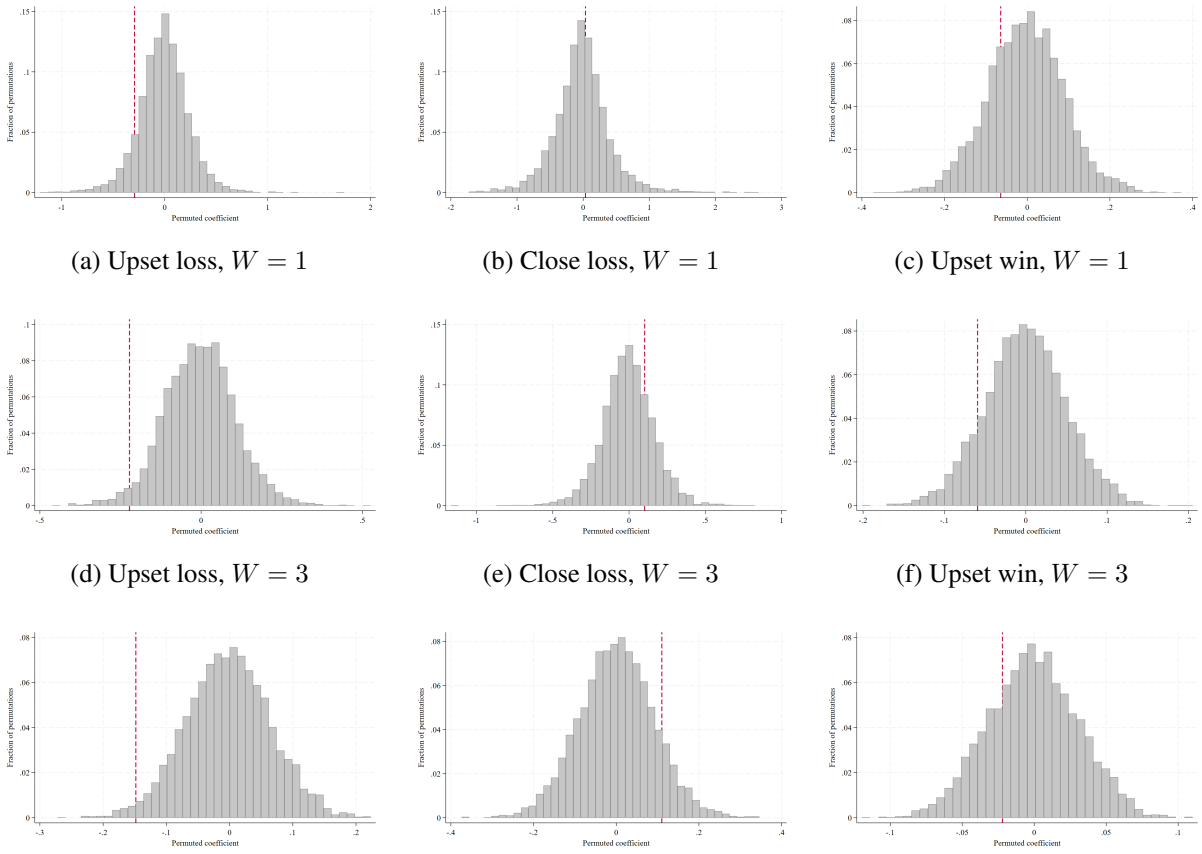
Figure 6: Robustness to alternative controls and fixed effects.

Notes: Dependent variable: *Non-interference: public criticism* (1–5, higher values indicating greater support for non-interference). Each marker is the upset-loss coefficient, with its 95% confidence interval, at the one-, three-, and seven-day windows; the baseline reproduces Table 3 and each later specification adds to it: interview-timing controls (day-of-week and month fixed effects, a holiday indicator); team-performance controls (home status, form, standing, importance, favorite probability, transfer spending, attendance); province-level economic controls under separate province and year fixed effects; and exact home-match attendance (Transfermarkt). Close-loss and upset-win placebos are omitted for clarity; confidence intervals use province-clustered standard errors; CGSS weights ($N = 35{,}103$). Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.2 Randomization Inference

The stability of the point estimate still leaves open the question of what to conclude from p-values that rest on few treated clusters. We assess statistical significance using randomization inference (Young, 2019; MacKinnon and Webb, 2020), permuting the full vector of match-outcome variables across province $\times$ date cells 5,000 times while holding covariates and fixed effects constant. The permutation is at the province $\times$ date level because that is the unit at which the treatment varies; the exercise therefore asks how often so few treated cells could produce estimates this large by chance, while the wild-cluster bootstrap of Appendix Table A1 addresses dependence at the broader province level, and the two procedures answer complementary questions. The p-value is the share of permutations producing a studentized coefficient (t-ratio) at least as extreme as observed in absolute value (two-sided).

Figure 7 plots the permutation distributions. The upset-loss coefficient falls in the tail of its distribution ($p = 0.099$, 0.006, and 0.034 in the one-, three-, and seven-day windows), while close-loss and upset-win coefficients sit near the center ($p = 0.833$ and 0.383 in the one-day window). A stricter variant that permutes the shock bundle only within province-by-year blocks, holding each province-year's realized exposure fixed and randomizing only its timing across interview dates, yields, as expected, larger p-values (0.254, 0.065, and 0.136); it is the most demanding test the design permits, and the three-day estimate remains at the margin under it. Under both schemes, the placebos remain central while upset losses sit in or near the left tail, a contrast that is difficult to attribute to chance alone.



(a) Upset loss, $W = 1$ (b) Close loss, $W = 1$ (c) Upset win, $W = 1$

(d) Upset loss, $W = 3$ (e) Close loss, $W = 3$ (f) Upset win, $W = 3$

(g) Upset loss, $W = 7$ (h) Close loss, $W = 7$ (i) Upset win, $W = 7$

Figure 7: Randomization inference: permutation distributions of coefficients.

Notes: Dependent variable: *Non-interference: public criticism* (1–5, higher values indicating greater support for non-interference). Each panel plots the coefficient distribution from 5,000 permutations of the match-outcome variables across province × date cells, with the red line at the actual estimate; rows are the 1-, 3-, and 7-day windows and columns the upset losses, close losses, and upset wins. All specifications include province × year fixed effects, individual, media, and Card predicted-category controls; CGSS weights ($N = 35{,}103$). The p-values reported in Section 6.2 are the share of permutations whose studentized coefficient is at least as extreme as observed in absolute value (two-sided; Young, 2019).

7 Conclusion

Unexpected football losses temporarily increase expressed acceptance of government control over public criticism. An upset loss reduces support for non-interference with public criticism by 0.293 points (0.26 standard deviations) in the one-day window. The effect fades over the following week and is absent for close losses and upset wins.

Prior work on emotional spillovers from sports finds that where these shocks reach political outcomes, the response follows the valence of the shock: favorable surprises have been argued to lift incumbent support and unfavorable ones erode trust in leaders, just as negative shocks produce violence, harsher sentencing, and stock market declines elsewhere. Our results show that this mapping reflects the outcome an emotion meets rather than a property of negative affect itself. Where the state is widely regarded as the guarantor of social order, transient negative arousal appears to increase demand for that order rather than erode trust in the state. This response is consistent with a compensatory-control interpretation (Kay et al., 2008) and with non-sports evidence showing that incidental emotions shift judgments toward what the emotion makes people seek: protectionist and redistributive policies under induced negative emotions (Algan et al., 2026), and the status quo under rain-induced negative affect (Meier et al., 2019). The direction of the spillover is thus set by the outcome margin the emotion encounters and by the state's position as the provider of order, not by the valence of the shock alone.

These boundaries are best read as scope conditions on this interpretation rather than as a map of regime types. The pro-control turn should surface where the state is a salient and legitimate provider of order (Kay et al., 2008, 2009); where a shaken sense of control has some other ready outlet, the same defeat need not raise acceptance of state limits on public criticism. It also presumes an emotionally charged shock: the outcome has to be widely followed and bound up with local identification, so in settings where the sport means something different or draws a thinner following, both the affective jolt

and its spillover should fade. The sign we recover reflects a low-control, targetless appraisal; where incidental feeling is coupled more loosely to political judgment, or where a clear object of blame is at hand, the reaction could weaken or even reverse into a same-valence one (Healy et al., 2010).

Political attitudes are often treated as stable dispositions, yet our results show that they respond, at the margin, to emotional cues unrelated to government performance. Standard accounts of trust in China emphasize rational evaluation of government output (Fang et al., 2025) or durable dispositions transmitted across generations (Booth et al., 2022). Zhang et al. (2025) establish a causal link from life satisfaction to political trust in China, and Zhang and Awaworyi Churchill (2020) show that income inequality erodes subjective well-being partly through neighborhood trust, but both turn on life satisfaction, a cognitive evaluation of life circumstances. Our results point to a more volatile component that moves with transient emotional states, and that may help explain short-run swings in public opinion otherwise difficult to attribute to policy changes.

Several limitations apply. First, our estimates are intent-to-treat effects at the province level; the CGSS has no direct information on fandom or match awareness. Second, what we capture is a short-run movement in expressed attitudes that attenuates over the week, not a lasting revision of underlying political preferences. Third, with so few treated clusters, statistical significance varies across inference procedures. Randomization inference clears the 5% threshold at three days under the baseline permutation scheme ($p = 0.006$) but is marginal under the stricter within-province-year scheme ($p = 0.065$), clears it at seven days under the baseline scheme only ($p = 0.034$ and 0.136), and is weakest in the one-day window, where upset losses are rarest; the wild-cluster bootstrap does not reach conventional thresholds, so significance should be read jointly with the specificity pattern (Section 6). Fourth, the CGSS is a repeated cross-section, so we cannot track within-individual changes or test whether repeated exposure produces cumulative effects.

Finally, we cannot fully separate a purely affective reaction from one partly mediated by associations with football mismanagement. Nor can we separate it from a response to the wave of public venting a defeat can set off, in which case respondents react to the criticism they see around them rather than to their own arousal. Likewise, a transient rise in self-censorship on a sensitive item (Robinson and Tannenberg, 2019) would be observationally equivalent to a change in underlying attitudes.

The contribution, however, does not hinge on which proximate channel operates: a spillover from an apolitical sporting outcome to expressed political attitudes is the phenomenon we document; the effect is most pronounced in tolerance of public criticism rather than reflecting a generalized demand for state control (Figure 3); and the treatment is the unexpected defeat of a provincial club rather than the national team, the usual focus of football-corruption discourse. Relatedly, although the surprise of the result points to a low-control, uncertainty appraisal (Smith and Ellsworth, 1985; Lerner and Keltner, 2001), supporters can also direct anger at players, coaches, or referees, and survey data cannot isolate which negative emotion (anxiety, frustration, disappointment, or anger at the team) carries the effect; direct measurement of affect could resolve this. The short-run malleability of political attitudes, orthogonal to what the state has done, is itself worth pursuing; future work with panel data or real-time survey designs could address these limitations.

Data availability

The replication package for this article is available in the China Economic Review Dataverse at https://doi.org/10.7910/DVN/FDTRJS. It contains all code, all non-restricted data, data documentation, and the logs and outputs of a complete run. The study is classified as Tier 2 because two data sources cannot be redistributed, even in processed form: the CGSS microdata and the Transfermarkt data. The README explains how to obtain the CGSS data. Researchers with authorized access to the restricted datasets can use the provided materials to reproduce all results reported in the article.

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Appendix

A. Inference with Few Treated Clusters

Table A1 collects the inference for the upset-loss coefficient given the few treated provinces: the analytic province-clustered p-value, the Webb wild-cluster bootstrap, and leave-one-province-out and leave-one-event-out estimates; randomization inference is reported in Section 6.2. The point estimate is robust to dropping any single province or event, while the wild-cluster bootstrap p-values are larger, as the null-imposing wild-cluster bootstrap is expected to under-reject with so few treated clusters (MacKinnon and Webb, 2018).

Table A1: Inference for the upset-loss effect with few treated clusters.

	Day 1	Day 3	Day 7
Upset-loss coefficient	-0.293	-0.222	-0.148
clustered standard error	(0.094)	(0.054)	(0.058)
analytic p-value	0.004	0.000	0.016
wild-cluster bootstrap p	0.204	0.146	0.234
Leave-one-province-out range	$[-0.43,\ -0.18]$	$[-0.24,\ -0.17]$	$[-0.18,\ -0.08]$
sign flips; worst p	$0;\ 0.004$	$0;\ 0.019$	$0;\ 0.205$
Leave-one-event-out range	$[-0.38,\ -0.18]$	$[-0.32,\ -0.15]$	$[-0.17,\ -0.11]$
sign flips; worst p	$0;\ 0.175$	$0;\ 0.049$	$0;\ 0.108$

Notes: Dependent variable: *Non-interference: public criticism* (1–5, higher values indicating greater support for non-interference); all entries are for the upset-loss coefficient under the Table 3 specification. Each column reports the coefficient with its clustered standard error, the analytic and Webb wild-cluster-bootstrap p-values, and leave-one-province-out and leave-one-event-out coefficient ranges (each treated province, respectively each treated province $\times$ date event, dropped in turn); sign flips counts the re-estimations with a positive coefficient. Football-data odds (Pinnacle closing), CGSS 2015–2023; province-clustered standard errors; CGSS sampling weights.

B. Additional Robustness

Appendix Table A2 adds the cumulative goal deficit of the upset losses, separating the occurrence of an upset from its severity.

Table A2: Upset-loss severity: occurrence versus margin of defeat.

	Day 1	Day 3	Day 7
Upset loss (count)	-0.485***	-0.602***	-0.273**
	(0.097)	(0.217)	(0.115)
Upset-loss goal deficit	0.090	0.183**	0.070
	(0.105)	(0.067)	(0.042)
Close loss	0.035	0.106	0.108*
	(0.091)	(0.136)	(0.062)
Upset win	-0.069	-0.073*	-0.030
	(0.065)	(0.040)	(0.039)
Mean of dep. var.	2.551	2.551	2.551
Province × year FE	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes
Media controls	Yes	Yes	Yes
N	35,103	35,103	35,103
R^2	0.057	0.058	0.058

Notes: Dependent variable: *Non-interference: public criticism* (1–5, higher values indicating greater support for non-interference). The specification adds the cumulative goal deficit of the upset losses (the margin of defeat) alongside the upset-loss count; close loss and upset win enter as counts. All specifications include province × year fixed effects, individual controls, media controls, and Card predicted-category controls; pre-interview window $[d - W,\, d - 1]$ with $W \in \{1, 3, 7\}$; province-clustered standard errors in parentheses; CGSS sampling weights ($N = 35{,}103$). $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$.

C. Heterogeneity

Table A3 reports formal interaction tests of heterogeneity by internet use, agricultural hukou, and gender. No upset-loss interaction is significant in all three windows simultaneously, so the heterogeneity is read as suggestive and underpowered rather than as identifying which respondents drive the effect. The two margins that do reach significance also point in different directions: high internet use amplifies the response in the shortest windows, while agricultural hukou amplifies it at the three- and seven-day horizons even though rural respondents are on average less connected. This tension is further reason not to read the heterogeneity as isolating a single exposure channel.

Table A3: Heterogeneity in the upset-loss effect.

	High internet use			Agricultural hukou			Female		
	Day 1	Day 3	Day 7	Day 1	Day 3	Day 7	Day 1	Day 3	Day 7
Upset loss	-0.134	-0.132**	-0.106***	-0.242	-0.148*	-0.087	-0.309**	-0.359***	-0.097
	(0.162)	(0.060)	(0.030)	(0.218)	(0.078)	(0.070)	(0.116)	(0.118)	(0.102)
Interaction	-0.326*	-0.180**	-0.079	-0.087	-0.217***	-0.244*	0.033	0.308	-0.111
	(0.180)	(0.082)	(0.093)	(0.273)	(0.060)	(0.124)	(0.065)	(0.192)	(0.121)
Close loss	0.060	0.129	0.149**	0.993**	0.452**	0.202**	0.328***	0.207	0.103
	(0.091)	(0.174)	(0.059)	(0.448)	(0.172)	(0.093)	(0.092)	(0.138)	(0.082)
Interaction	-0.376***	-0.108	-0.114	-1.117**	-0.609***	-0.167	-0.598***	-0.225***	0.015
	(0.033)	(0.196)	(0.200)	(0.447)	(0.075)	(0.110)	(0.045)	(0.038)	(0.083)
Upset win	-0.001	-0.027	-0.009	-0.131	-0.097	-0.024	-0.116	-0.105*	-0.024
	(0.106)	(0.044)	(0.049)	(0.126)	(0.071)	(0.044)	(0.090)	(0.052)	(0.056)
Interaction	-0.153	-0.087	-0.036	0.107	0.052	-0.006	0.102	0.083*	0.005
	(0.121)	(0.065)	(0.073)	(0.162)	(0.072)	(0.036)	(0.108)	(0.047)	(0.049)
Province × year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Media controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	35,103	35,103	35,103	35,103	35,103	35,103	35,103	35,103	35,103
R^2	0.057	0.058	0.058	0.057	0.058	0.058	0.057	0.058	0.058

Notes: Dependent variable: *Non-interference: public criticism* (1–5, higher values indicating greater support for non-interference). Each column is one full-sample regression interacting every shock with the block's subgroup indicator (high internet use, agricultural hukou, or female); the indented Interaction row is the coefficient on the interaction and the shock row is the effect in the omitted category, so the subgroup's effect is the sum of the two. All specifications include province × year fixed effects, individual controls, media controls, and Card predicted-category controls; pre-interview window $[d-W,\ d-1]$ with $W \in \{1, 3, 7\}$; province-clustered standard errors in parentheses; CGSS sampling weights ($N = 35{,}103$). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D. Ordered Logit and Probit Estimates

Table A4 re-estimates the main specification with ordered logit and ordered probit, replacing the absorbed province-by-year fixed effects with indicators. The upset-loss coefficient is negative and significant in every window under both models, so the results of Table 3 do not depend on treating the five-point outcome as linear.

Table A4: Ordered logit and ordered probit estimates of the upset-loss effect.

	Day 1	Day 3	Day 7
Ordered logit	-0.416***	-0.358***	-0.237***
	(0.157)	(0.093)	(0.082)
Ordered probit	-0.274***	-0.210***	-0.125**
	(0.073)	(0.050)	(0.053)
Province × year indicators	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes
Media controls	Yes	Yes	Yes
N	35,103	35,103	35,103

Notes: Dependent variable: *Non-interference: public criticism* (1–5, higher values indicating greater support for non-interference). Each cell reports the upset-loss coefficient from an ordered logit or ordered probit re-estimation of the Table 3 specification on its estimation sample, with the absorbed province × year fixed effects replaced by province × year indicators. All specifications include individual controls, media controls, and Card predicted-category controls; pre-interview window $[d - W,\ d - 1]$ with $W \in \{1, 3, 7\}$; province-clustered standard errors in parentheses; CGSS sampling weights ($N = 35{,}103$). $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$.